B. E–VI Sem- Theme Based Project (2025-26)

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ABSTRACT

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Title of the Project:

DESIGN OF A SCALABLE KALMAN FILTER ARCHITECTURE USING SYSTEMVERILOG

ABSTRACT:

Kalman filters are widely used for optimal state estimation in noisy environments, including sensor processing, navigation, and tracking systems. This project presents the RTL design of a parameterizable Kalman filter architecture in SystemVerilog, derived from core Kalman equations and implemented using a modular, scalable approach. The design includes configurable blocks for state prediction, covariance update, Kalman gain computation, and measurement update, enabling easy extension from single-state to multi-state estimation. Fixed-point arithmetic is adopted to optimize accuracy, area, and performance, making the architecture suitable as a reusable IP core for real-time signal processing applications.

AIM & OBJECTIVES:

- To derive the Kalman filter mathematical model and translate it into a hardware-oriented RTL architecture.
- To design modular RTL blocks for state prediction, error covariance update, Kalman gain computation, and measurement update.
- To implement fixed-point arithmetic units optimized for hardware efficiency while maintaining acceptable estimation accuracy.
- To ensure synthesizability and hardware readiness by adhering to RTL coding guidelines and evaluating area, timing, and resource utilization.

TOOLS:

- Synopsys VCS – For RTL functional simulation and basic sanity checks of the design • MATLAB – For Kalman filter algorithm modelling and fixed-point behaviour analysis
- Synopsys Design Compiler – For RTL synthesis, area, timing, and power analysis

DESIGN METHODOLOGY:

- Algorithm Understanding and Requirement Definition
- Algorithm Modeling and Fixed-Point Analysis
- Architecture Design and Block Partitioning
- RTL Design Using System Verilog
- Design-Level Simulation and Debug
- Synthesis and Optimization

CO-PO/PSO MAPPING

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3			2							2	3	2	
CO2	3	3	3		3							2	3	2	
CO3	2		3	2	3				2			2	3	3	
CO4	2		2		3					2		3	2	2	
CO5									3	3	3				

1.
2.
3.
4.

Supervisor

Student(s)